Artificial Intelligence

Network Architectures and Logic Gates

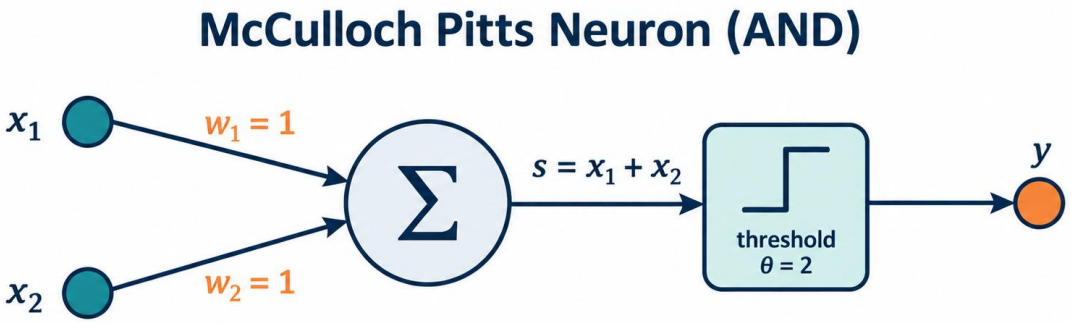


Lecture 05

The Islamic University in Najaf Al-Ashraf
College of Technical Engineering
Department of Computer Techniques Engineering
Instructor Jaafar Al-Kafnawiz

Najaf Al-Ashraf

Threshold neuron and logic gates



AND Truth Table

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

A weighted sum and a threshold map binary inputs to a gate output.

Architecture components

Input layer receives features.

Hidden layers transform representations.

Output layer produces the prediction.

Connections carry weighted signals.

Bias terms shift activation thresholds.

Single-layer feedforward network

Inputs connect directly to output neurons.

There is no trainable hidden layer.

The network represents linear decision boundaries.

Perceptron and Adeline use this structure.

Multilayer feedforward network

One or more hidden layers transform the inputs.

Nonlinear activations support nonlinear mappings.

Signals move from input to output without cycles.

Backpropagation trains the weights.

Recurrent and competitive networks

Recurrent networks carry an internal state through feedback connections.

They suit sequences in which earlier inputs affect later outputs.

Competitive networks compare neuron responses and select a winner.

The winner, or its neighborhood, learns.

This supports clustering and self-organization.

Logic gates as neuron models

Binary inputs encode False as 0 and True as 1.

Weights represent the contribution of each input.

A threshold determines the binary output.

Simple gates illustrate neural computation.

AND gate

Desired output is 1 only when both inputs are 1.

Choose w1 = 1, w2 = 1, threshold θ = 2.

Net equals x1 + x2.

Output is 1 when net ≥ 2. $y = \text{step}(x_1 + x_2 - 2)$.

OR and NOT gates

OR: $y = \text{step}(x_1 + x_2 - 1)$, with $\text{step}(z) = 1$ for $z \geq 0$.

For OR, only input (0, 0) produces output 0.

NOT: $y = \text{step}(-x + 0.5)$.

The negative weight reverses the input.

For NOT, input 0 gives output 1, and input 1 gives output 0.

NAND and NOR gates

NAND is the inverse of AND.

NOR is the inverse of OR.

Negative weights and a positive bias implement inhibition.

NAND alone can construct all Boolean functions.

The need for hidden units in XOR

XOR outputs 1 when the inputs differ.

Positive examples occupy opposite corners.

No single line separates the two classes.

A hidden layer combines multiple linear regions.

A single threshold neuron cannot implement XOR.

Feedforward and recurrent comparison

Feedforward: signals pass toward the output without cycles.

A standard feedforward model has no state carried from previous samples.

Recurrent: feedback carries information across time steps.

Sequence order matters, and training must account for temporal dependencies.

Threshold-neuron equation

Net input: net = w1x1 + w2x2 + b.

Output: y = 1 when net is at least 0.

Weights and bias determine the decision boundary.

Logic gate verification

List every input combination in the truth table.

Compute net = $w_1x_1 + w_2x_2 + b$ for each row.

Apply the specified threshold, including the equality case.

Compare every result with the desired gate output.

Architecture selection

Use a single layer for linearly separable classes.

Use hidden layers when one boundary is insufficient.

Use recurrent connections when order and memory matter.

Architecture benefits and limitations

Single layer: simple computation, but only linear decision boundaries.

Hidden layers: nonlinear representations, with more parameters and training work.

Recurrent connections: sequence memory, with more difficult temporal training.

Competitive learning: useful grouping, but some neurons may rarely win.

Network architectures: comparison

Aspect	Feedforward	Recurrent
Connections	No directed cycles	Feedback connections
Memory	No carried internal state	State across time steps
Typical task	Fixed input-output map	Sequence processing